ISSSP 2026 — TRACK 3: LUNAR SURFACE ROBOTICS · FINALS SCORING CRITERION 2

Official Competition Poster Specification & Blueprint

Complete Scientific Poster Copy, Technical Schemes, 3-Column Architectural Layout, and Artistry Design Guidelines (Deadline: 31st Oct — Max Score: 10/10)

Poster Title: L-MINER Mk-II Autonomous Science Payload	**Format:** ISO A0 Portrait (841 x 1189 mm) / 300 DPI
Scoring Item: Item 2: Poster (10 Pts)	**Authors:** Haider, Wasay, Saad, Hajra (IST)

1. Poster Design Specifications & Scoring Mandate

According to Table 6 (Finals Scoring Rules of Track 3), the Poster deliverable (due 31st October) carries 10 points. Evaluators score on: 'Project scheme display, Include Team Information, Background Introduction, Research Content And Methods, etc.' It also directly anchors the 5-point 'Artistry' score in the Finals.

- **Physical Dimensions:** Standard ISO A0 Portrait (841 mm width x 1189 mm height; 33.1 x 46.8 inches).
- **Digital Canvas:** 9933 x 14043 pixels at 300 DPI (CMYK Color Space for professional high-contrast printing).
- **Grid System:** 3-Column Symmetric Grid with 30 mm outer margins and 20 mm inter-column gutters.
- **Visual Flow:** Header -> Column 1 (Context & Autonomy) -> Column 2 (Sensors & ML) -> Column 3 (HUD & ISRU).

2. Poster Header & Team Information Block

POSTER TITLE (96 pt Bold): L-MINER Mk-II: Autonomous Multi-Tier Payload for Subsurface Lunar Volatile Prospecting and Mineral Classification
SUBTITLE (44 pt Medium): Dual-Modality Subsurface GPR and VNIR-SWIR Spectrometry with Edge Machine Learning for Lunar South Pole Cold Traps
AUTHORS (32 pt): Haider Taqvee[1], Abdul Wasay[1], Saad[1], Hajra[1]
AFFILIATION (24 pt): [1] Department of Space Science & Aeronautics, Institute of Space Technology (IST), Islamabad, Pakistan
ISSSP 2026 — Track 3: Lunar Surface Robotics and ISRU Exploration | Project ID: LMINER-IST-2026

3. Print-Ready Text Copy Across All Poster Panels

Panel 1: Background & Scientific Problem

The Lunar South Pole features Permanently Shadowed Regions (PSRs) where temperatures remain below 40 K. Orbital datasets confirm surface frost, but cannot resolve subsurface stratigraphy or volume.
The Ground-Truth Gap: Optical cameras only sample the top 10 microns of dust. Earth teleoperation is impossible due to communication shadows and 3-second roundtrip delays.
Objective: Autonomous rover fusing subsurface radar sounding (0–3 m) with optical spectroscopy (0.5–2.5 μm) and edge ML.

Panel 2: Hierarchical Autonomy (L0 / L1 / L2)

• **L0 Safety Reflex (10 Hz):** Real-time obstacle bypass using HazCam point-clouds. Blocks drive into slopes >15° or hazard score >0.75.
• **L1 Resource Manager (1 Hz):** Monitors battery SoC and electronics temperatures. Automatically pauses scans at SoC < 30% or T > 50°C for solar recharge.
• **L2 Science Prospecting (On Demand):** Executes multi-physics sensor sounding, feature extraction, and ML classification at target coordinates.

Panel 3: Multi-Physics Sensor Modeling

Synthetic sensor modeling grounded in Apollo sample returns and USGS splib07 library:
• **500 MHz GPR:** 128 x 256 B-scan profiles across 0–3 m depth via dielectric permittivity contrast (Ice: A=0.85, Ilmenite: A=0.55, Pyroxene: A=0.65).
• **VNIR-SWIR Spectrometer (0.5–2.5 μm):** Active illumination in dark PSRs. Captures diagnostic bands: Water Ice (1.25, 1.5, 2.0 μm), Pyroxene (1.0 & 2.0 μm), Ilmenite (flat R~0.07).

Panel 4: Edge Feature Extraction & ML Engine

• **3,300x Edge Data Reduction:** Compresses 32,968 raw floats into a 10-D feature vector (4 GPR features + 6 spectral features).
• **Random Forest Classifier (100 Trees):** 95.8% overall multiclass accuracy (Ice: 99.2%, Ilm: 96.5%, Pyr: 94.8%, Reg: 92.5%).
• **Sub-15 ms Inference Latency** on radiation-hardened ARM Cortex-R5 / RTG4 FPGA processors.

Panel 5: Interactive 3D Digital Twin & Mission HUD

• **3D Digital Twin:** High-density 96 x 96 crater terrain mesh with 3° polar lighting, 6-wheel rocker-bogie rover, terrain slope adaptation, and subterranean deposit ellipsoids sized by ML confidence.
• **Streamlit Mission Control:** Real-time telemetry, dual GPR B-scan + A-scan energy plots, USGS benchmark overlay, and survey controls.

Panel 6: Quantitative ISRU Yields & Mission Impact

• **Survey Yield:** 7 water ice deposits localized at avg 0.9 m depth across 60 x 60 m survey zone.
• **Resource Potential:** ~57.8 Tonnes H■O mapped (1500 kg/m³ density, 5.5 wt% ice).
• **Propellant Output:** ~51.4 Tonnes LOX + ~6.4 Tonnes LH2, capable of refueling two lunar ascent return vehicles.

4. Poster Layout Wireframe & Graphic Design Blueprint

Column 1 (247 mm)	Column 2 (247 mm)	Column 3 (247 mm)
Panel 1: Context & Gap • Lunar South Pole Cold Traps • In-situ prospecting challenge	**Panel 3: Sensor Physics** • 500 MHz GPR profile • VNIR-SWIR absorption curves	**Panel 5: 3D Digital Twin** • Rocker-bogie 3D rover mesh • Subsurface deposit spheres
Panel 2: Autonomy • L0 Reflex, L1 Resource, L2 ML • Flowchart of state transitions	**Panel 4: Machine Learning** • 10-D Feature extraction • Confusion matrix & accuracy	**Panel 6: ISRU Potential** • 57.8 Tonnes H■O calculation • LOX/LH2 propellant yield